

**PRAGATI ENGINEERING COLLEGE: SURAMPALEM
(AUTONOMOUS)**

III B.Tech I Semester Regular Examinations, November - 2025

**EDGE COMPUTING
(CSE Honors)**

Time: 3 hours

Max. Marks: 70

Note:

- i. Question No. 1 shall contain 10 compulsory short answer questions (2 questions from each unit) for a total of 20 marks such that each question carries 2 marks.
- ii. In each of the questions from 2 to the last question, there shall be either/or type questions of 10 marks each.

Q. No.	Question	BTL	CO	Marks
1.	a) What is Latency?	K1	CO1	2M
	b) Define Edge Computing	K1	CO1	2M
	c) Define an edge gateway.	K1	CO2	2M
	d) Expand LPWAN.	K1	CO2	2M
	e) What is Google Coral?	K1	CO3	2M
	f) What is TensorFlow Lite?	K1	CO3	2M
	g) Define data aggregation.	K1	CO4	2M
	h) What is secure boot?	K1	CO4	2M
	i) Define smart grid.	K1	CO5	2M
	j) What is 6G-enabled edge?	K1	CO5	2M
UNIT – I				
2.	a) Explain the role of edge computing in smart cities.	K2	CO1	5M
	b) Discuss the major benefits of edge computing.	K2	CO1	5M
OR				
3.	a) Compare Cloud, Fog, and Edge computing with suitable examples.	K4	CO1	5M
	b) Demonstrate the applications of edge computing in autonomous systems.	K3	CO1	5M
UNIT – II				
4.	a) What are the differences among MQTT, CoAP, and AMQP?.	K2	CO2	5M
	b) Explain the importance of network protocols in edge computing.	K2	CO2	5M
OR				
5.	a) Demonstrate and sketch the Edge Computing architecture and its key components.	K2	CO2	5M
	b) Discuss HTTP/2 and GRPC communication in edge environments.	K2	CO2	5M
UNIT – III				

6.	a)	Explain the components and usage of Google Edge TPU.	K2	CO3	5M
	b)	What are the comparisons between Raspberry Pi, NVIDIA?	K2	CO3	5M
OR					
7.	a)	Discuss the features of AWS Greengrass and Azure IoT Edge.	K2	CO3	5M
	b)	Demonstrate the AZURE IOT Edge.	K3	CO3	5M
UNIT – IV					
8.	a)	Explain the secure boot and trusted execution environments.	K2	CO4	5M
	b)	Describe privacy-preserving analytics with examples.	K2	CO4	5M
OR					
9.	a)	Describe federated learning and its role in secure edge AI.	K2	CO4	5M
	b)	Explain real-time data stream processing techniques at the edge.	K2	CO4	5M
UNIT – V					
10.	a)	Discuss the concept of Edge-to-Cloud continuum.	K2	CO5	5M
	b)	Explain the use of edge computing in AR/VR applications.	K2	CO5	5M
OR					
11.	a)	Illustrate the importance of edge computing for Industry 4.0 applications.	K2	CO5	5M
	b)	Analyse the impact of 6G on the future of edge computing.	K4	CO5	5M